

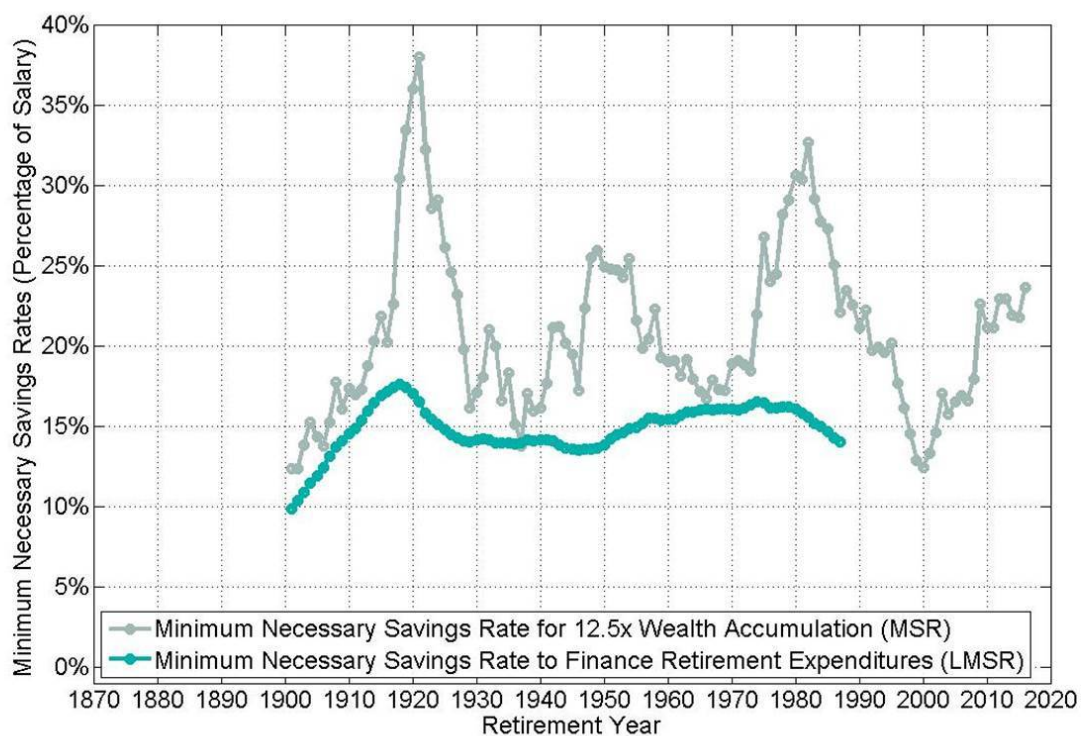


Exhibit 5.11 provides these results, showing both the savings rate needed to accumulate 12.5 times final salary (the MSR described above) and the life-cycle-based minimum savings rate (LMSR) needed to finance desired expenditures. The LMSR curve provides my idea about the “safe savings rates.” From rolling historical periods, they represent the minimum necessary savings rate for the accumulation phase to pay for the desired retirement expenditures. In the context of Bengen’s original study, the maximum value of the LMSR curve (17.6 percent in 1918) becomes the SAFEMIN savings rate from a lifetime perspective that corresponds to Bengen’s SAFEMAX withdrawal rate. Had the baseline individual used a fixed 17.6 percent savings rate, she would have always saved enough to finance her desired retirement expenses, having barely accomplished this in the worst-case retirement year of 1918.

Retirement planning in the context of the LMSR curve is less prone to making large sacrifices in order to follow a conservative strategy. In the context of safe withdrawal rates, if someone used a 4 percent withdrawal rate at a time that would have supported an 8 percent withdrawal rate, she is